

# Fourier transform questions

## Exercise 1.

Answer the following questions:

How many samples do the signal consist of?

$$N = 1024$$

What is the sampling frequency?

$$F_s = 1024 \text{ Hz}$$

What is the sampling time for each sample?

$$t_s = \frac{1}{F_s} = \frac{1}{1024} \text{ s}$$

What is the total sampling time?

$$T = \frac{N}{F_s} = 1 \text{ s} \quad (\text{if the frequency } f \text{ is given in units of } \text{Hz} = \frac{1}{\text{s}})$$

What is the frequency resolution in spectrum?

$$\Delta f = \frac{1}{T} = 1 \text{ Hz}$$

What is the highest frequency in spectrum?

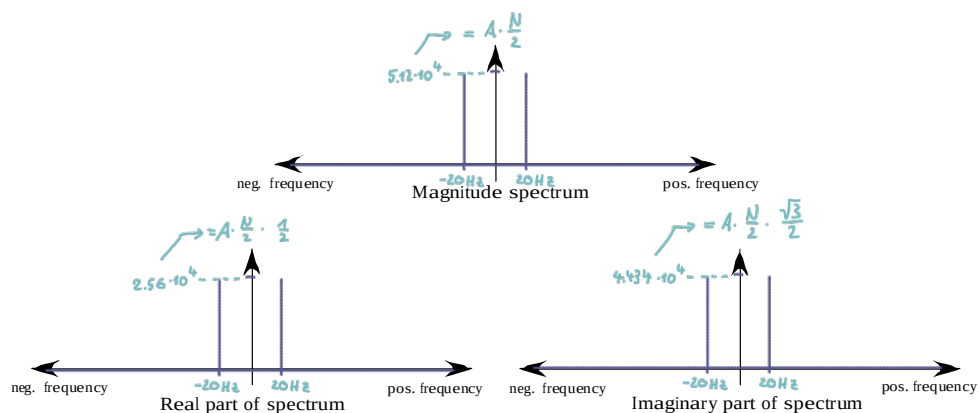
$$F_{\max} = \frac{F_s}{2} = 512 \text{ Hz}$$

MatLab makes use of the FFT function when calculating a Fourier transform if the number of samples is an even power of 2. Otherwise the transform is calculated using DFT (Direct Fourier Transform), which takes much longer. Below are the formulas MatLab uses for DFT. Fill in the missing parts of a DFT calculation from the first exercise

$$F_x(k) = \sum_{n=0}^{N-1} x(n) \cdot e^{-i(2\pi k \frac{n}{N})}$$

$$x(n) = \frac{1}{N} \cdot \sum_{k=0}^{N-1} F_x(k) \cdot e^{i(2\pi n \frac{k}{N})}$$

Mark in the diagram the spectrum of the transformed signal in exercise 1. Map the amplitude and frequency, and don't forget the negative frequencies.



Assume that we have a complex vector which includes the Fourier transform of a real signal. How do you plot the signal?

The problem is to calculate the time scale, then make the inverse transform, and finally plot the right part of the complex inverse transform against the time scale.

Consider that we have a spectrum with  $N = 1000$  frequencies (positive + negative). The frequency resolution is  $2 \text{ Hz} = \Delta f$   $\Delta f = \frac{1}{\Delta t}$ ?

What is the sampling time,  $\Delta t$ ?

$$\Delta t = \frac{T}{N} = 0.5 \text{ ms}$$

What is the sampling frequency,  $f_s$ ?

$$f_s = \frac{1}{\Delta t} = 2000 \text{ Hz}$$

How long time  $T$  is the signal sampled?

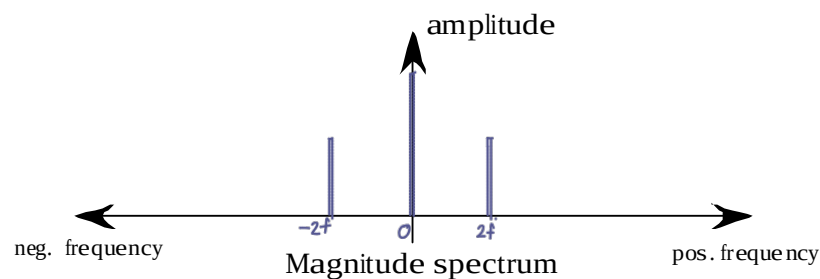
$$T = \frac{1}{\Delta f} = 0.5 \text{ s}$$

Write down some MatLab commands that generates a time vector, calculates the inverse transform, and plots the signal.

```
% given: spectrum Fx
N=1000; df=2;
T=1/df; % total time of sampling
dt=T/N; % time per sample
t=dt*(0:N-1) % time vector
x=ifft(Fx); % inverse FFT
plot(t,x)
```

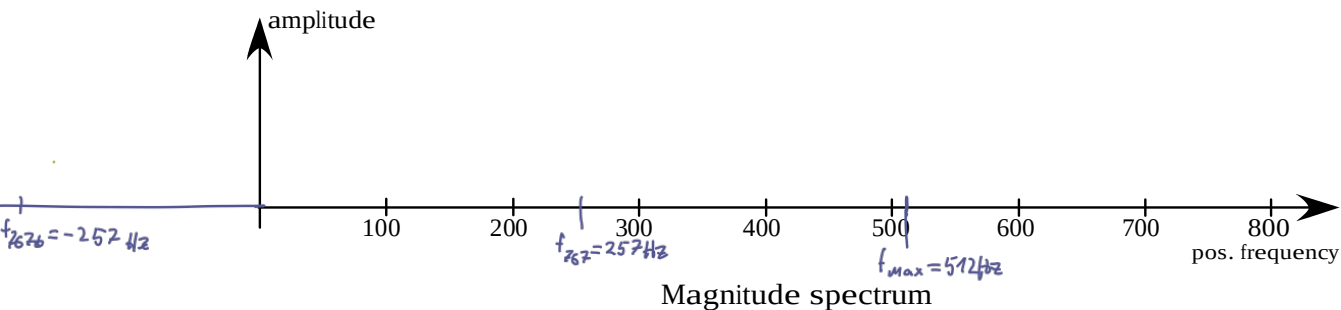
## Exercise 2:

Plot the spectrum of the squared sinusoid



### Exercise 3:

Mark the maximal frequency in the spectrum (Nyquist frequency). The sampling frequency is 1024 Hz.



Mark the aliased frequencies obtained when you are sampling and transforming 767 Hz.

Where can you find the sampled frequencies  $f_1 = 400$  Hz,  $f_2 = 1200$  Hz, and  $f_3 = 2000$  Hz. For sampled signals, the spectrum is periodic. Are the aliased frequencies positive or negative? Can you show this? You can make use of the plots shown later which indicate how it is possible to represent a spectrum of a sinusoidal signal with a frequency  $f$ .

The following aliased frequencies are observed when we sample with  $f_s = 1024$  Hz

| $F_{in}$ | $F_{out}$ |
|----------|-----------|
| 767 Hz   | -257 Hz   |
| 400 Hz   | 400 Hz    |
| 1200 Hz  | 176 Hz    |
| 2000 Hz  | -424 Hz   |

Which frequencies are negative? Study the real and imaginary parts and the phase

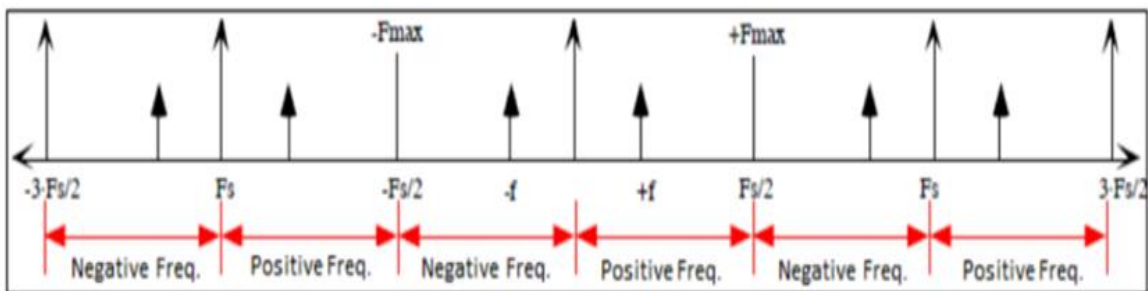
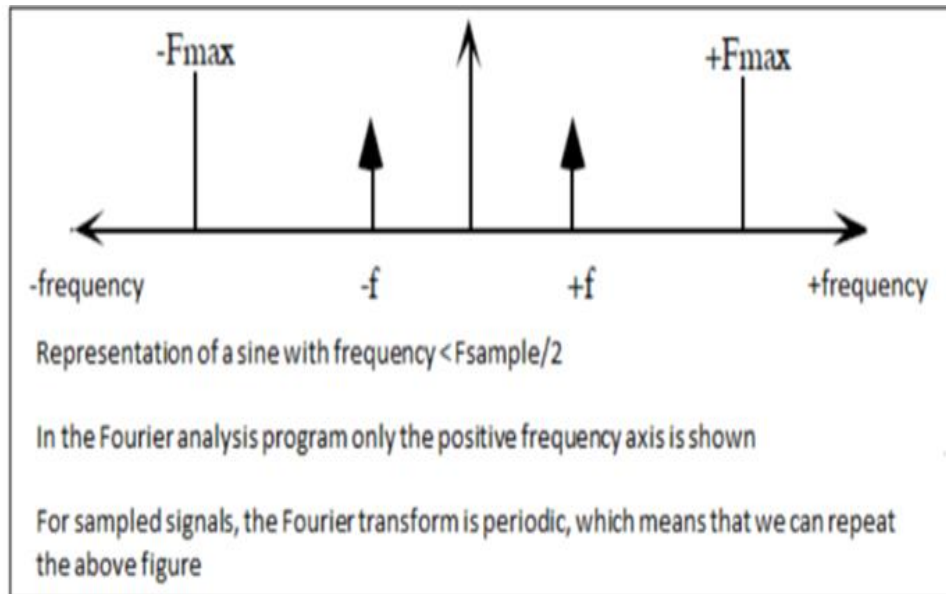
### Exercise 4:

Describe the difference between the transform of a sinusoid with and without a window function. In the exercise the suggested window was a Hanning window a sinusoid of 100.5 Hz.

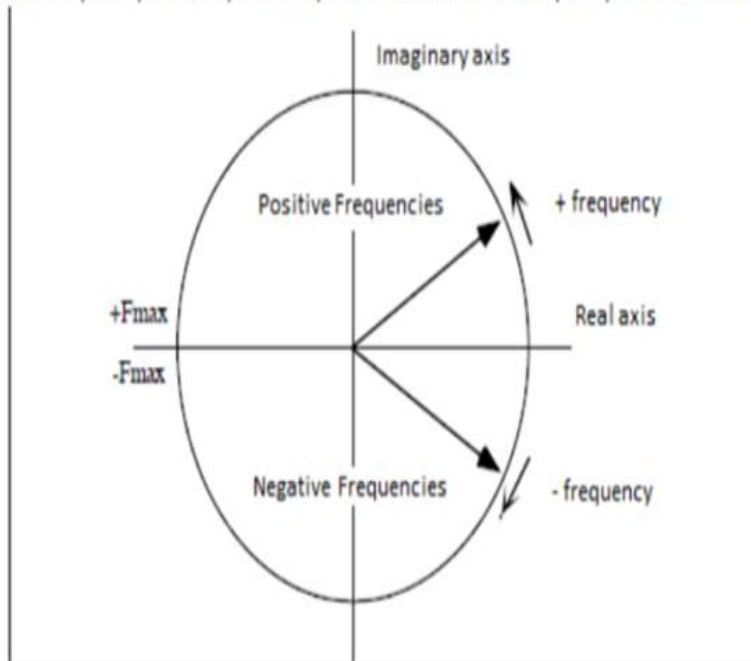
Without a window function, the Fourier transform has peaks at  $-100.5$  Hz and  $100.5$  Hz which spread out toward nearby frequencies. Even at far away frequencies, the spectrum never reaches 0. Multiplying the sinusoid by a Hanning window, the peaks become much narrower and the spectrum quickly goes close to 0 away from the peaks. The amplitudes of the peaks are also significantly lower since the Hanning window only contains values  $\leq 1$  and thus lowers the overall amplitude of the sinusoid.

Why does the spectrum broaden for a sinusoidal signal with a non-integer frequency, for example, 67.2 Hz? Assume the sampling rate is still 1024 Hz.

We have  $N = 1024$  samples at a sampling rate  $f_s = 1024$  Hz, so the total time is  $T = \frac{N}{f_s} = 1$  s. Any signal with a frequency that is not an integer multiple of  $\frac{1}{T} = 1$  Hz will not have completed a full cycle at the end of 1 s. When periodically repeating such a signal, there will therefore be discontinuities which add additional (high) frequencies to the spectrum.



Another way to represent a periodic spectrum is to draw the frequency axis round a circle



### Exercise 5:

What are the frequency components of a symmetric pulse train with a period of 100 ms?



period = 100 ms  $\Rightarrow f = 10\text{ Hz}$ . The frequency components are  $10\text{ Hz} + n \cdot 20\text{ Hz}$ ;  $n = 0, 1, 2, \dots$ .  
The corresponding magnitudes are  $\frac{651.85}{1+2n}$ . This is mirrored for negative frequencies.

Describe the spectrum of a pulse train with a duty cycle of 25% and a period of 125 ms? Compare with the corresponding symmetric pulse train, and comment.

period = 125 ms  $\Rightarrow f = 8\text{ Hz}$ . The spectrum contains  $8\text{ Hz} + n \cdot 16\text{ Hz}$ ;  $n = 0, 1, 2, \dots$   
similarity to the 50% duty cycle, but also  $16\text{ Hz} + n \cdot 32\text{ Hz}$ .  
It has periodic bumps since the components spaced by 32 Hz have higher amplitude than those spaced by 16 Hz. Additionally, the spectrum contains a peak at 0 Hz.  
The corresponding symmetric pulse only has  $8\text{ Hz} + n \cdot 16\text{ Hz}$ , so it only contains odd multiples of the fundamental frequency.

What is the relationship between the harmonics and the fundamental frequency of a square wave?

We call the fundamental frequency  $f_0$ . Then the square wave contains frequencies  $f_n = (2n+1)f_0$  (and mirrored for negative frequencies) which we call the  $n$ -th harmonics. The  $n$ -th harmonic has magnitude  $A_n = \frac{A_0}{2n+1}$  where  $A_0$  is the amplitude of the fundamental frequency.

Explain where the aliased frequencies end up when the fundamental frequency is 128 Hz.

| $F_{in}/\text{Hz}$ | $F_{out}/\text{Hz}$ | $F_{in}/\text{Hz}$ | $F_{out}/\text{Hz}$ |
|--------------------|---------------------|--------------------|---------------------|
| 128                | 128                 | -128               | -128                |
| 384                | 384                 | -384               | -384                |
| 640                | -384                | -640               | 384                 |
| 896                | -128                | -896               | 128                 |
| $\vdots$           |                     |                    |                     |

The aliased frequencies overlap with non-aliased frequencies meaning that their amplitudes add up. (This can also reduce the resulting amplitude since the phases don't match)

What is the relationship between the harmonics and the fundamental frequency of a triangular wave?

the  $n$ -th harmonic is  $f_n = (n+1) \cdot f_0$  and its magnitude is  $A_n = \frac{A_0}{n+1}$ .

### Exercise 6:

Describe the frequency spectrum of a unit pulse (a signal with only one sample different from zero).

The frequency spectrum of a unit pulse is constant (in modulus) over all frequencies and has the same amplitude as the pulse. The argument is not constant (unless the pulse is at  $t=0$ ) and is different depending on when the pulse happens.

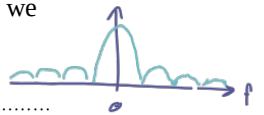
Describe the spectrum of two single pulses separated by a time,  $\Delta t$ . Can you calculate a mathematical expression of the transform?

The magnitude spectrum  $|F_x(f)|$  is periodic, containing bumps of equal amplitude. The farther apart the pulses, the more frequent the bumps become.

Since  $F[\delta(t-t_0)] = \int dt \exp(-i \cdot 2\pi f t) \delta(t-t_0) = \exp(-i \cdot 2\pi f t_0)$ , the FT of two pulses (at  $t_1, t_2$ ) is  $\exp(-i \cdot 2\pi f t_1) + \exp(-i \cdot 2\pi f t_2)$ .

Describe the magnitude spectrum of a single pulse with a width  $\Delta t$ . Describe what happens when we increase or decrease  $\Delta t$ .

The spectrum has repeating bumps that decrease in magnitude away from  $f=0$ . The bumps become narrower with increasing  $\Delta t$ , repeating more often, such that the spectrum always contains  $m_2 - m_1$  bumps ( $m_1$  is the start and  $m_2$  the end of the region rel. to 1 in the original signal).



### Exercise 7:

Which frequencies are included in the spectrum of a carrier wave of 100 Hz modulated with a square wave of 10 Hz. The modulation index = 0.5. Give the 5 frequencies with the greatest amplitude.

The 5 frequencies with the highest magnitude (in descending order of amplitude): 100 Hz, 90 Hz, 110 Hz, 80 Hz, 120 Hz. The spectrum goes on like this in steps of 20 Hz, getting lower in amplitude with increasing distance to the peak at 100 Hz.

### Exercise 8:

Investigate how large a signal/noise ratio it is possible to have to be able to determine a sinusoidal signal. Show the ratio in dB.

We see that the power density of white noise of amplitude 1 (i.e. going from -0.5 to 0.5) is  $\frac{1}{12}$ , whereas it is  $\frac{1}{2}$  for  $\sin(\dots)$ . So we multiply the noise by  $\sqrt{\frac{6}{1}}$  where  $\eta$  is the SNR and  $\sqrt{6}$  is there to equalize the power density to that of the sine. We try out different SNR values and see that down to about  $\eta = 0.023$  we can still reliably see the peak of the sine's frequency in the spectrum,  $\eta = 0.023$  corresponds to  $\eta = -32.8$  dB.

### Exercise 9:

Fill in the frequencies of the touch-tone keypad.

$$N = 2800$$

$$F_s = 8000 \text{ Hz}$$

$$\Delta f = \frac{F_s}{N} = 2.86 \text{ Hz}$$

Column frequencies

|                 |               |                |                |                |
|-----------------|---------------|----------------|----------------|----------------|
|                 |               | <u>1209 Hz</u> | <u>1337 Hz</u> | <u>1427 Hz</u> |
|                 | <u>657 Hz</u> | 1              | 2              | 3              |
|                 | <u>771 Hz</u> | 4              | 5              | 6              |
|                 | <u>851 Hz</u> | 7              | 8              | 9              |
|                 | <u>940 Hz</u> | *              | 0              | #              |
| Row frequencies |               |                |                |                |

What can you expect for accuracy if one takes all the samples from the signals that define the different numbers?

$$\Delta f = \dots\dots\dots \text{ Hz}$$

Three waves available for each frequency

$$\rightarrow \bar{f} = \frac{f_1 + f_2 + f_3}{3}$$

$$\Delta \bar{f} = \frac{\sigma_f}{\sqrt{3}} = \frac{2.86 \text{ Hz}}{\sqrt{3}} = 1.65 \text{ Hz}$$